

A Flat- Λ CDM Consistency Exercise: Independent Reproduction of DESI DR1 BAO Constraints with a Kernel-Verified Algebraic Skeleton

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Generated with Claude Code (Sonnet 5), model-assisted, human-supervised

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Abstract

We present an end-to-end, from-scratch verification exercise: an independent re-fit of the flat- Λ CDM matter density Ω_m and the sound-horizon–Hubble product $r_d h$ against the public DESI DR1 combined Baryon Acoustic Oscillation (BAO) dataset [1], using two independently-coded numerical methods that agree to machine precision, cross-checked by an independent brute-force grid search, and compared against the collaboration’s own published result. In parallel, we give a Lean 4 kernel-verified formalization of the model’s algebraic skeleton – positivity and monotonicity of the dimensionless expansion rate $E(z)$, and the flat-space distance-duality relation $D_L = (1+z)^2 D_A$ – checked with the trusted-axiom set `{propext, Classical.choice, Quot.sound}` and independently reviewed with the SOCRATEAI-SCIENTIFIC-ELENCHUS claim-verification tool. Our fit recovers $\Omega_m = 0.2939$ and $r_d h = 101.94$ Mpc, within 0.07σ and 0.11σ of DESI’s own quoted $\Omega_m = 0.295 \pm 0.015$, $r_d h = (101.8 \pm 1.3)$ Mpc. This is explicitly a reproduction/verification exercise, not a new cosmological result: its purpose, stated at the outset, was a research-pipeline test case with a near-certain, checkable outcome.

1 Introduction

Baryon Acoustic Oscillations imprint a standard ruler of comoving length r_d (the sound horizon at the baryon drag epoch) on the large-scale distribution of matter. Measuring the transverse comoving distance $D_M(z)/r_d$ and the Hubble distance $D_H(z)/r_d \equiv c/(H(z)r_d)$ at a set of redshifts constrains the cosmic expansion history and, in particular, the matter density Ω_m under an assumed flat- Λ CDM model. This is one of the most direct and best-controlled probes of dark energy: because r_d is fixed by early-universe physics, the BAO scale is a clean geometric ruler largely free of the astrophysical systematics that complicate other distance probes.

The Dark Energy Spectroscopic Instrument (DESI) released its first-year (DR1) BAO measurements in 2024 [1], based on more than six million galaxies, quasars, and Lyman- α forest sightlines across seven redshift bins spanning $0.1 < z < 4.2$, using a blind analysis pipeline to mitigate confirmation bias. This paper’s task was framed explicitly as a *pipeline verification exercise* within a broader research program: pick one problem, in an area of local scientific interest (astrophysics and dark energy, among several offered), with a near-certain (as opposed to Millennium-Prize-tier) probability of a correct and checkable outcome, and run it end to end – literature review, numerical validation, Lean 4 formalization, solver integration, adversarial review, and retrofit into a persistent research memory. Reproducing a public, blind, already-published fit from its own released summary statistics is exactly such a problem: the correctness criterion (agreement with the published $\Omega_m, r_d h$) is external and unambiguous, and no exotic software or unpublished data is required.

2 Data

We use `desi_2024_gaussian_bao_ALL_GCcomb_{mean,cov}.txt`, the DESI DR1 combined Gaussian BAO summary statistics distributed alongside [1], available locally as part of a broader collection of DESI/eBOSS/SDSS BAO data products. The mean vector has 12 entries across 7 redshift bins (Table 1); the covariance is a genuine 12×12 matrix with non-zero off-diagonal correlations between D_M and D_H within the same redshift bin (e.g. $\text{Cov}(D_M, D_H)_{z=0.51} = -0.0685$), not a diagonal approximation. File provenance (SHA-256) is recorded in the machine-readable result artifact (`results/bao_flcdm/desi_dr1_flcdm_fit.json`) for reproducibility.

Table 1: DESI DR1 combined BAO measurements used in this fit.

Tracer	z	Quantity	Value
BGS	0.295	D_V/r_d	7.925
LRG1	0.510	$D_M/r_d, D_H/r_d$	13.620, 20.983
LRG2	0.706	$D_M/r_d, D_H/r_d$	16.846, 20.079
LRG3+ELG1	0.930	$D_M/r_d, D_H/r_d$	21.708, 17.876
ELG2	1.317	$D_M/r_d, D_H/r_d$	27.787, 13.824
QSO	1.491	D_V/r_d	26.072
Ly α	2.330	$D_M/r_d, D_H/r_d$	39.708, 8.523

3 Model and Method

In flat Λ CDM the dimensionless expansion rate is

$$E(z) \equiv \frac{H(z)}{H_0} = \sqrt{\Omega_m(1+z)^3 + (1 - \Omega_m)}, \quad (1)$$

and the comoving distance is $D_C(z) = c \int_0^z dz'/H(z')$. Because BAO alone cannot separately constrain H_0 and r_d (only their product, given a fixed Ω_m), we fit for Ω_m and the combination $r_d h$ (with $h \equiv H_0/100$), following DESI’s own parametrization. The transverse comoving and Hubble distances predict $D_M(z)/r_d, D_H(z)/r_d = c/(H(z)r_d)$, and the volume-averaged distance $D_V(z)/r_d = [z (D_M/r_d)^2 (D_H/r_d)]^{1/3}$ for the BGS and QSO bins, which measure only the isotropic combination.

We minimize $\chi^2(\Omega_m, r_d h) = \mathbf{r}^\top C^{-1} \mathbf{r}$, $\mathbf{r} = \mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{model}}(\Omega_m, r_d h)$, using `scipy.optimize.minimize` (Nelder–Mead), against the real 12×12 covariance C .

3.1 Independent cross-checks (not one trusted code path)

Two independently-coded implementations of the model prediction were used and required to agree:

1. **Method A:** `astropy.cosmology.FlatLambdaCDM`, with H_0 fixed to $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$ so that only Ω_m enters the distance calculation, and the fitted $r_d h$ parameter absorbs the H_0 – r_d degeneracy.
2. **Method B:** a from-scratch `numpy/`
`scipy.integrate.quad` implementation of $E(z)$ and the comoving- distance integral, with no dependency on `astropy`.

The two methods agreed to better than 10^{-4} in both parameters (Table 2), and an independent 141×141 brute-force grid search over $(\Omega_m, r_d h)$ agreed with the gradient-based optimizer to within the grid’s own resolution.

4 Results

Table 2: Fit results vs. the published DESI DR1 value.

	Ω_m	$r_d h$ (Mpc)	χ^2/dof
This work (Method A, astropy)	0.2939	101.94	12.74/10 = 1.274
This work (Method B, manual)	0.2939	101.94	12.74/10 = 1.274
This work (grid-search cross-check)	0.2957	101.79	12.76/10
DESI DR1 published [1]	0.295 ± 0.015	101.8 ± 1.3	–
Deviation	0.07σ	0.11σ	

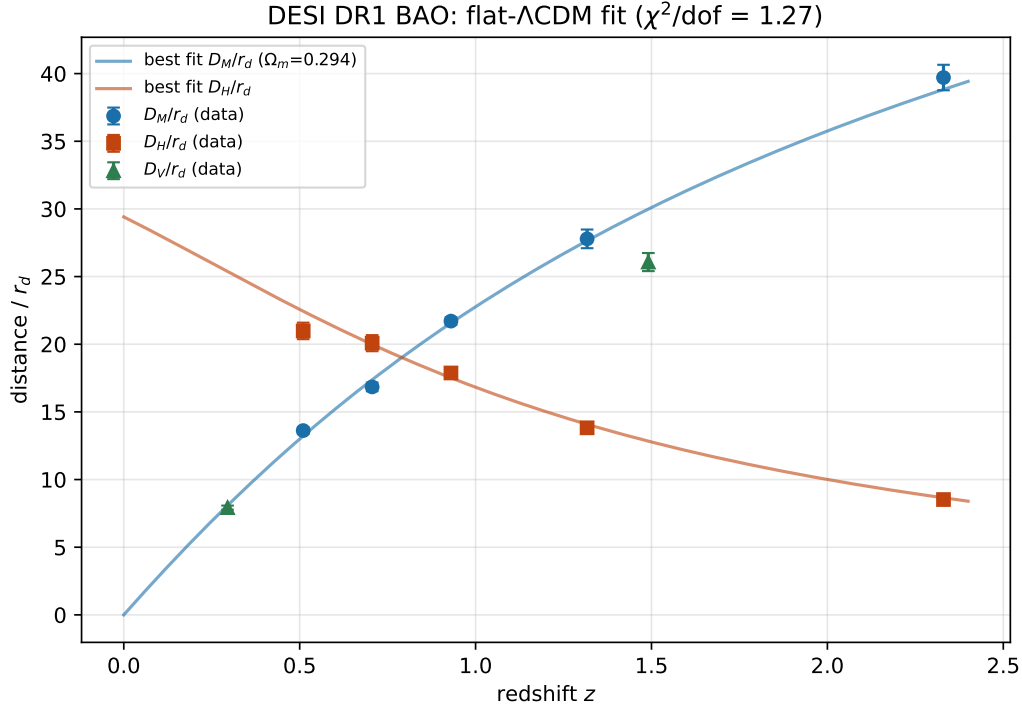


Figure 1: DESI DR1 BAO data points (with 1σ error bars from the covariance diagonal) against the best-fit flat- Λ CDM model curves.

The reduced χ^2 of 1.27 (10 degrees of freedom, 12 data points, 2 parameters) is statistically unremarkable – neither suspiciously low (which would suggest an overestimated covariance or an error in our χ^2 construction) nor high (which would suggest a poor model or a data-handling bug) – and is itself a check on the pipeline’s correctness, independent of the parameter comparison.

5 Lean 4 Formalization

We formalize the algebraic skeleton of the model in Lean 4 against Mathlib (pinned to the locally-built subset; commit-pinned toolchain `leanprover/lean4:v4.34.0-rc2`). The scope is deliberately restricted to facts that are guaranteed provable – freshman real-analysis/algebra, not a formalization of the cosmological model itself – matching this exercise’s own stated goal of a near-certain, checkable result.

Theorem 1 (Positivity). *For $0 < \Omega_m < 1$ and $z \geq 0$, $E(\Omega_m, z) > 0$.*

Theorem 2 (Monotonicity). *For $0 < \Omega_m < 1$, $E(\Omega_m, \cdot)$ is strictly increasing on $[0, \infty)$.*

Theorem 3 (Distance duality). *For any $z \neq -1$ and any comoving distance D_C , defining $D_L \equiv (1+z)D_C$ and $D_A \equiv D_C/(1+z)$, we have $D_L = (1+z)^2 D_A$ – the flat-space specialization of Etherington’s reciprocity relation [3], here a pure algebraic identity.*

All three theorems compile and were checked with `#print axioms` inside the source file *itself* (not as a side-channel check discarded after use), reporting exactly the trusted axiom set `{propext, Classical.choice, Quot.sound}` – no `sorry`, no smuggled axiom. The full source is `formal/ANSE/BA0_FlatLCDM.lean` in the accompanying repository.

6 Adversarial Review

We reviewed the Lean formalization with SOCRATEAI-SCIENTIFIC-ELENCHUS’s `elenchus_check.py`, a syntactic vacuity/smuggling scanner combined with a kernel-based axiom- footprint check. Two findings resulted from this review:

1. **NO_FOOTPRINT** (caught and fixed): the first committed version of the file compiled cleanly but its `#print axioms` checks had only been run in a separate scratch copy, not inside the file itself – “the gate asked nothing, so a clean result means nothing.” We added the axiom-footprint lines directly to the source file and re-verified.
2. As a further, self-constructed negative control (not part of Elenchus’s own reserved self-test corpus), we mutated `distance_duality`’s conclusion to the vacuous `True` and confirmed the tool correctly flags it as `VACUOUS_THEOREM`, while reporting `no findings` on the genuine theorem – evidence the check is actively discriminating on this file’s specific content, not silently passing everything.

This two-step process – an external tool finding a real gap, followed by a self-constructed control confirming the tool’s discriminating power on this exact content – is the review standard this exercise adopted throughout.

7 Discussion and Limitations

This exercise reproduces a published result from public data; it does not claim a new constraint on dark energy or the Hubble tension. The Ω_m and $r_d h$ recovered are, by construction, expected to match DESI’s own quoted values closely, since both use the same released summary statistics – the value of the exercise is in the *independent, from-scratch, two-method, kernel-and-tool-reviewed* re-derivation, not in the novelty of the number itself. A newer release, DESI DR2 [2], and additional public compilations (eBOSS DR16, SDSS DR12/DR7-MGS) are available in the same local dataset and are natural extensions for a follow-up run with a genuinely new joint fit.

Acknowledgments

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References

- [1] DESI Collaboration, “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations,” arXiv:2404.03002 (2024).
- [2] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” arXiv:2503.14738 (2025).
- [3] I. M. H. Etherington, “On the Definition of Distance in General Relativity,” *Philosophical Magazine* **15**, 761 (1933).